

Total No. of Questions : 8]

SEAT No. :

P7555

[Total No. of Pages : 2

[6180] - 66

T.E. (AI & DS)

PATTERN RECOGNITION

(2019 Pattern) (Semester - I) (317522 B) (Elective - I)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Identify the different examples of String Generation as Pattern Description? **[9]**

b) What are the different types of String Grammar? Explain it? **[8]**

OR

Q2) a) Compare Grammar and Language with suitable example? **[9]**

b) Describe the Grammar Based Approach and its uses in detail? **[8]**

Q3) a) Clarify canonical definite finite state grammar (CDFSG) with suitable example? **[9]**

b) Identify Isomorphism if we are having two Graph G1 and G2, each with p nodes? **[8]**

OR

Q4) a) Examine various applications of Relational Graph to Pattern Recognition? **[8]**

b) Determine the Recursive procedure to find Cliques with suitable example? **[9]**

P.T.O.

- Q5) a)** CAM and other Neural Memory Structure? **[9]**
b) Draw and Explain Artificial Neuron Activation and output Characteristics? **[9]**

OR

- Q6) a)** Explain Neural Networks as a Black Box Approach? **[9]**
b) Describe the different reasons to adopt a Neural Computational Architecture? **[9]**

- Q7) a)** Draw & Explain Summary of the Back Propagation learning Procedure? **[9]**
b) Clarify how the character classification is done with Pattern Associator? **[9]**

OR

- Q8) a)** Draw & Explain how to train the feedforward network using Generalized delta Rule? **[9]**
b) Describe the structure of a Multiple Layer Feedforward Network? **[9]**

